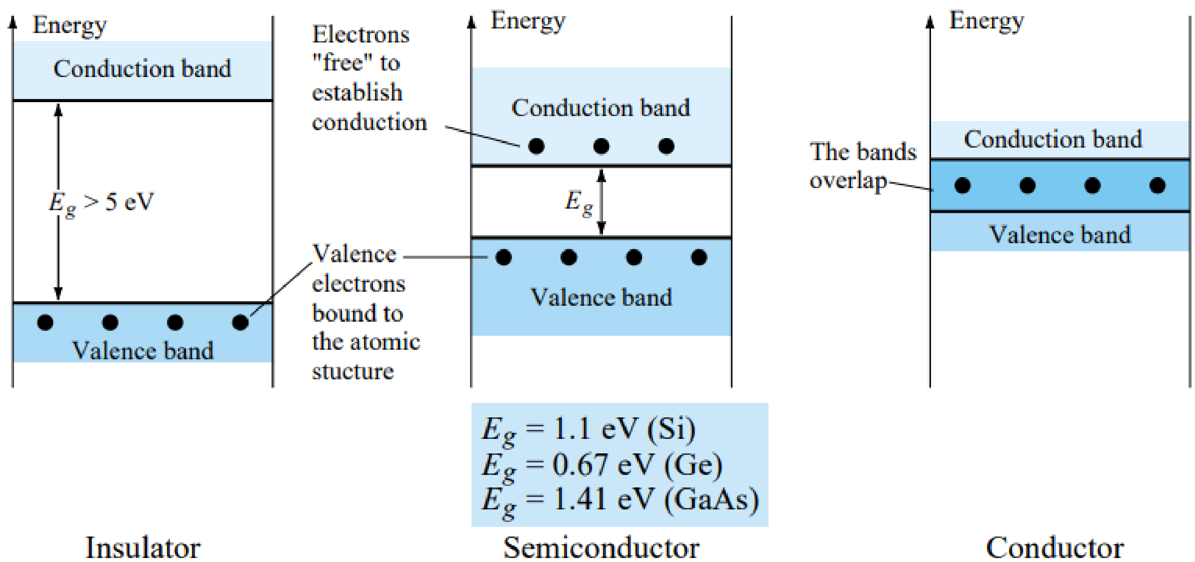
1.3 SEMICONDUCTOR MATERIALS

The label *semiconductor* itself provides a hint as to its characteristics. The prefix *semi-* is normally applied to a range of levels midway between two limits.

The term conductor is applied to any material that will support a generous flow of charge when a voltage source of limited magnitude is applied across its terminals.

An insulator is a material that offers a very low level of conductivity under pressure from an applied voltage source.

A semiconductor, therefore, is a material that has a conductivity level somewhere between the extremes of an insulator and a conductor.



1.5 EXTRINSIC MATERIALS— *n*- AND *p*-TYPE

The characteristics of semiconductor materials can be altered significantly by the addition of certain impurity atoms into the relatively pure semiconductor material. These impurities, although only added to perhaps 1 part in 10 million, can alter the band structure sufficiently to totally change the electrical properties of the material.

A semiconductor material that has been subjected to the doping process is called an extrinsic material.

n-Type Material

Both the *n*- and *p*-type materials are formed by adding a predetermined number of impurity atoms into a germanium or silicon base. The *n*-type is created by introducing those impurity elements that have *five* valence electrons (*pentavalent*), such as *antimony, arsenic,* and *phosphorus.*

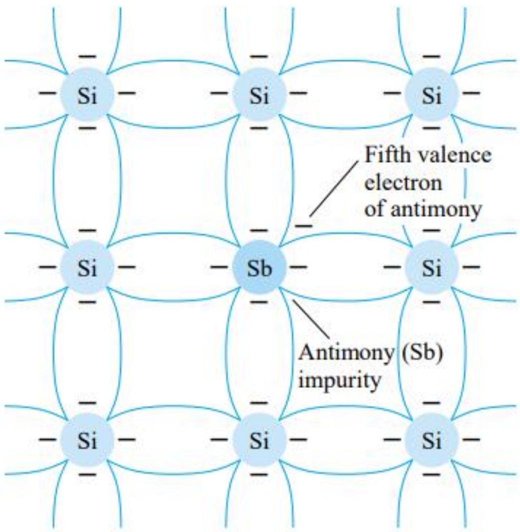


Figure 1.9 Antimony impurity in *n*-type material.

p-Type Material

The *p*-type material is formed by doping a pure germanium or silicon crystal with impurity atoms having *three* valence electrons. The elements most frequently used for this purpose are *boron, gallium,* and *indium.* The effect of one of these elements, boron, on a base of silicon is indicated in Fig. 1.11.

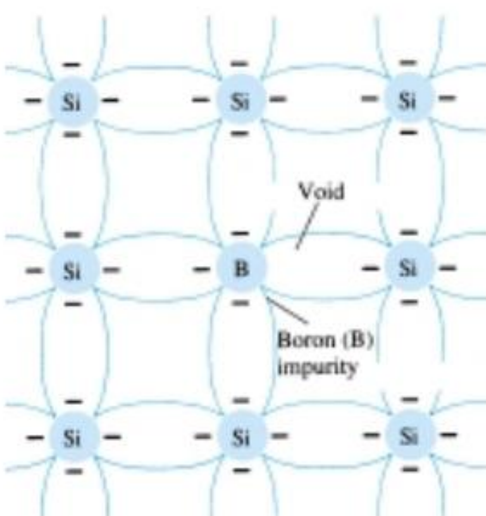


Figure 1.11 Boron impurity in *p*-type material.

Majority and Minority Carriers

In an n-type material (Fig. 1.13a) the electron is called the majority carrier and the hole the minority carrier.

For the *p*-type material the number of holes far outweighs the number of electrons, as shown in Fig. 1.13b. Therefore:

In a p-type material the hole is the majority carrier and the electron is the minority carrier.

When the fifth electron of a donor atom leaves the parent atom, the atom remaining acquires a net positive charge: hence the positive sign in the donor-ion representation. For similar reasons, the negative sign appears in the acceptor ion.

The *n*- and *p*-type materials represent the basic building blocks of semiconductor devices. We will find in the next section that the "joining" of a single *n*-type material with a *p*-type material will result in a semiconductor element of considerable importance in electronic systems.

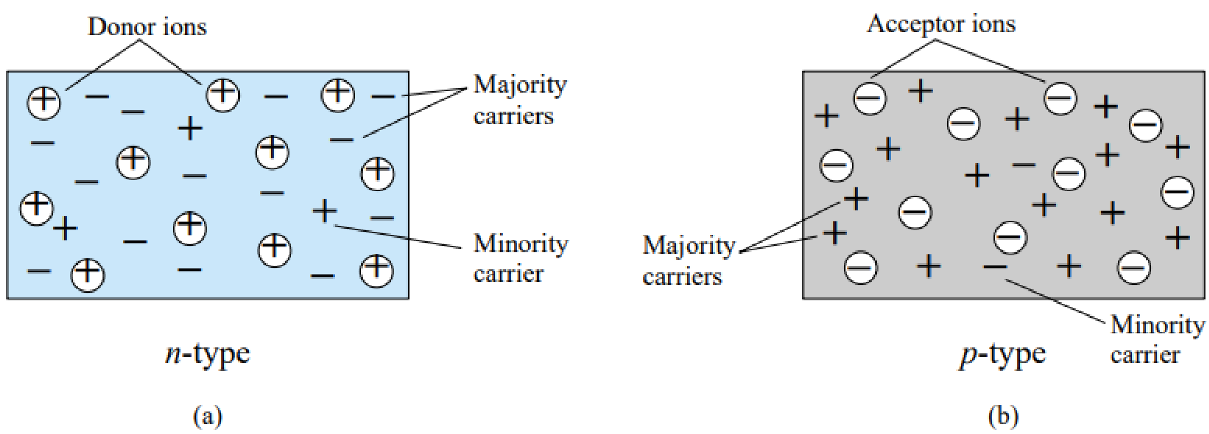


Figure 1.13 (a) *n*-type material; (b) *p*-type material.

1.6 SEMICONDUCTOR DIODE

In Section 1.5 both the *n*- and *p*-type materials were introduced. The semiconductor diode is formed by simply bringing these materials together (constructed from the same base—Ge or Si), as shown in Fig. 1.14, using techniques to be described in Chapter 20. At the instant the two materials are "joined" the electrons and holes in the region of the junction will combine, resulting in a lack of carriers in the region near the junction.

This region of uncovered positive and negative ions is called the depletion region due to the depletion of carriers in this region.

Since the diode is a two-terminal device, the application of a voltage across its terminals leaves three possibilities: *no bias* ($V_D = 0$ V), *forward bias* ($V_D > 0$ V), and *reverse bias* ($V_D < 0$ V). Each is a condition that will result in a response that the user must clearly understand if the device is to be applied effectively.

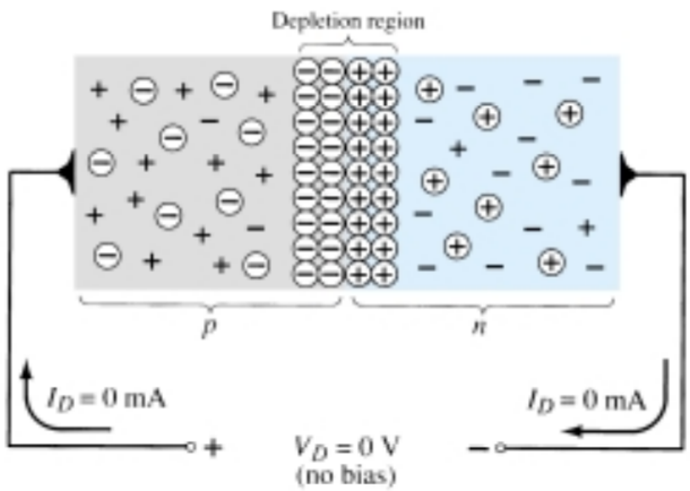


Figure 1.14 *p-n* junction with no external bias.

No Applied Bias (V_D = 0 V)

Under no-bias (no applied voltage) conditions, any minority carriers (holes) in the *n*-type material that find themselves within the depletion region will pass directly into the *p*-type material. The closer the minority carrier is to the junction, the greater the attraction for the layer of negative ions and the less the opposition of the positive ions in the depletion region of the *n*-type material. For the purposes of future discussions we shall assume that all the minority carriers of the *n*-type material that find themselves in the depletion region due to their random motion will pass directly into the *p*-type material. Similar discussion can be applied to the minority carriers (electrons) of the *p*-type material. This carrier flow has been indicated in Fig. 1.14 for the minority carriers of each material.

The majority carriers (electrons) of the *n*-type material must overcome the attractive forces of the layer of positive ions in the *n*-type material and the shield of negative ions in the *p*-type material to migrate into the area beyond the depletion region of the *p*-type material. However, the number of majority carriers is so large in the *n*-type material that there will invariably be a small number of majority carriers with sufficient kinetic energy to pass through the depletion region into the *p*-type material. Again, the same type of discussion can be applied to the majority carriers (holes) of the *p*-type material. The resulting flow due to the majority carriers is also shown in Fig. 1.14.

In the absence of an applied bias voltage, the net flow of charge in any one direction for a semiconductor diode is zero.

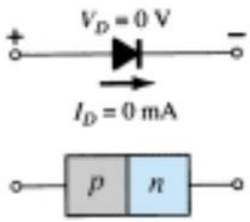


Figure 1.15 No-bias conditions for a semiconductor diode.

Reverse-Bias Condition ($V_D < 0$ V)

If an external potential of *V* volts is applied across the *p-n* junction such that the positive terminal is connected to the *n*-type material and the negative terminal is connected to the *p*-type material as shown in Fig. 1.16, the number of uncovered positive ions in the depletion region of the *n*-type material will increase due to the large number of "free" electrons drawn to the positive potential of the applied voltage. For similar reasons, the number of uncovered negative ions will increase in the *p*-type material. The net effect, therefore, is a widening of the depletion region. This widening of the depletion region will establish too great a barrier for the majority carriers to overcome, effectively reducing the majority carrier flow to zero as shown in Fig. 1.16.

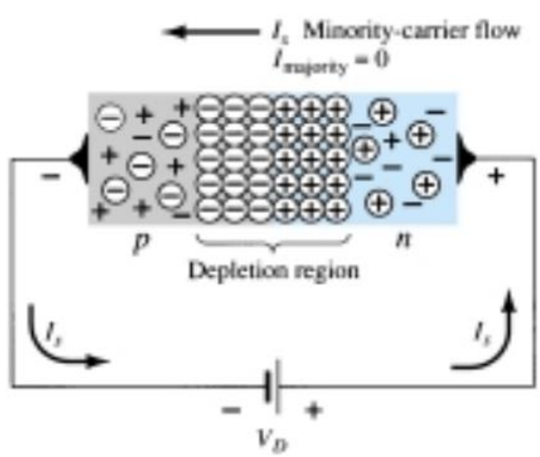


Figure 1.16 Reverse-biased *p-n* junction.

The number of minority carriers, however, that find themselves entering the depletion region will not change, resulting in minority-carrier flow vectors of the same magnitude indicated in Fig. 1.14 with no applied voltage.

The current that exists under reverse-bias conditions is called the reverse saturation current and is represented by I_s.

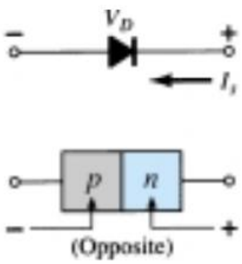


Figure 1.17 Reverse-bias conditions for a semiconductor diode.

Forward-Bias Condition ($V_D > 0$ V)

A *forward-bias* or "on" condition is established by applying the positive potential to the *p*-type material and the negative potential to the *n*-type material as shown in Fig. 1.18. For future reference, therefore:

A semiconductor diode is forward-biased when the association p-type *and positive and* n-type *and negative has been established.*

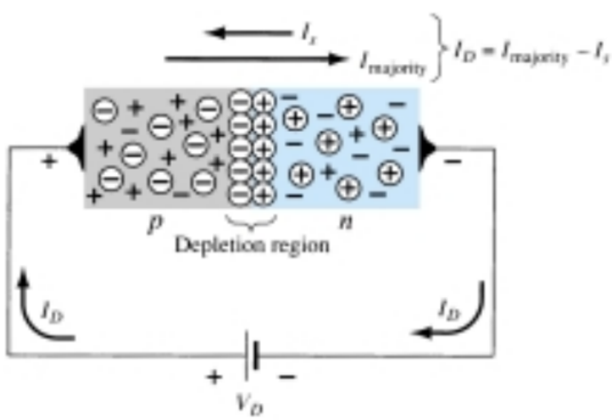


Figure 1.18 Forward-biased *p-n* junction.

The application of a forward-bias potential V_D will "pressure" electrons in the *n*-type material and holes in the *p*-type material to recombine with the ions near the boundary and reduce the width of the depletion region as shown in Fig. 1.18. The resulting minority-carrier flow of electrons from the *p*-type material to the *n*-type material (and of holes from the *n*-type material to the *p*-type material) has not changed in magnitude (since the conduction level is controlled primarily by the limited number of impurities in the material), but the reduction in the width of the depletion region has resulted in a heavy majority flow across the junction. An electron of the *n*-type material now "sees" a reduced barrier at the junction due to the reduced depletion region and a strong attraction for the positive potential applied to the *p*-type material. As the applied bias increases in magnitude the depletion region will continue to decrease in width until a flood of electrons can pass through the junction, re-

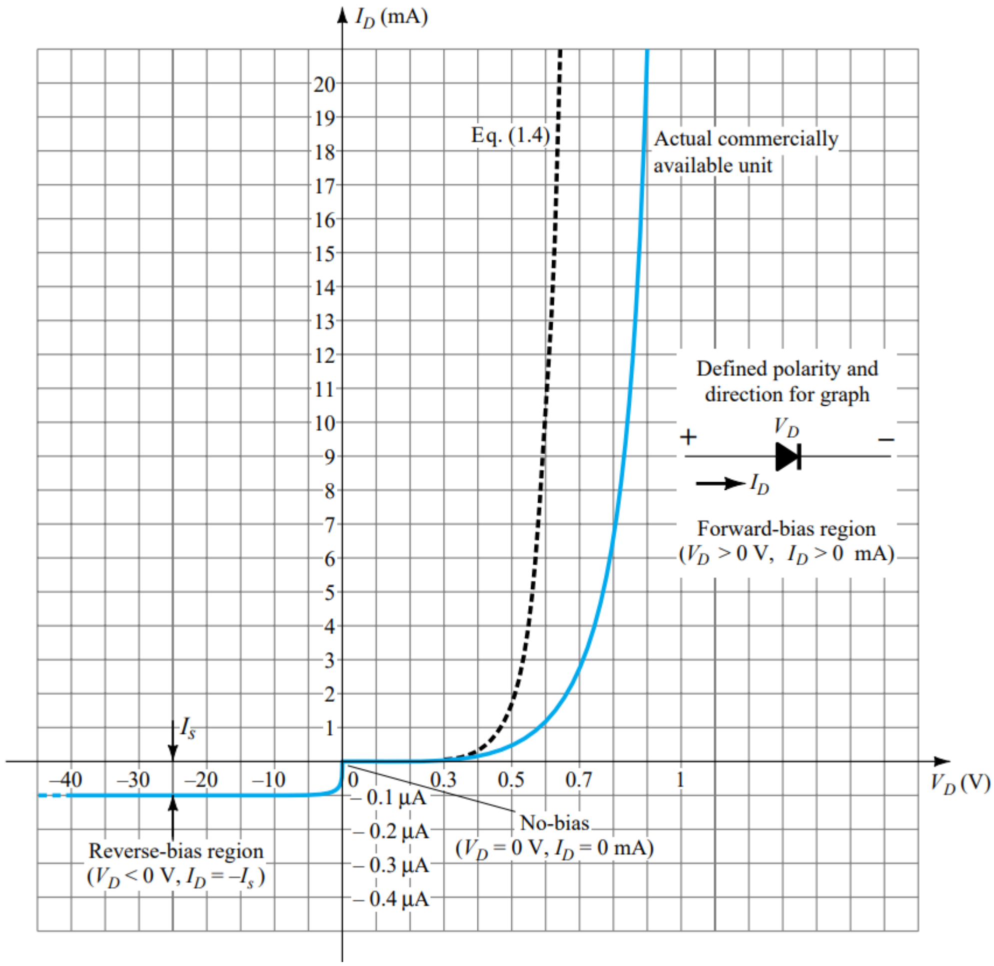


Figure 1.19 Silicon semiconductor diode characteristics.

sulting in an exponential rise in current as shown in the forward-bias region of the characteristics of Fig. 1.19. Note that the vertical scale of Fig. 1.19 is measured in milliamperes (although some semiconductor diodes will have a vertical scale measured in amperes) and the horizontal scale in the forward-bias region has a maximum of 1 V. Typically, therefore, the voltage across a forward-biased diode will be less than 1 V. Note also, how quickly the current rises beyond the knee of the curve.

It can be demonstrated through the use of solid-state physics that the general characteristics of a semiconductor diode can be defined by the following equation for the forward- and reverse-bias regions:

$$\boxed{I_D = I_s(e^{kV_D/T_K} - 1)} \tag{1.4}$$

where I_s = reverse saturation current

$k = 11{,}600/\eta$ with $\eta = 1$ for Ge and $\eta = 2$ for Si for relatively low levels of diode current (at or below the knee of the curve) and $\eta = 1$ for Ge and Si for higher levels of diode current (in the rapidly increasing section of the curve)

$T_K = T_C + 273°$

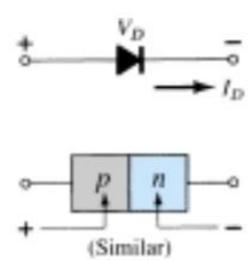


Figure 1.21 Forward-bias conditions for a semiconductor diode.

1.14 ZENER DIODES

The Zener region of Fig. 1.47 was discussed in some detail in Section 1.6. The characteristic drops in an almost vertical manner at a reverse-bias potential denoted V_Z. The fact that the curve drops down and away from the horizontal axis rather than up and away for the positive V_D region reveals that the current in the Zener region has a direction opposite to that of a forward-biased diode.

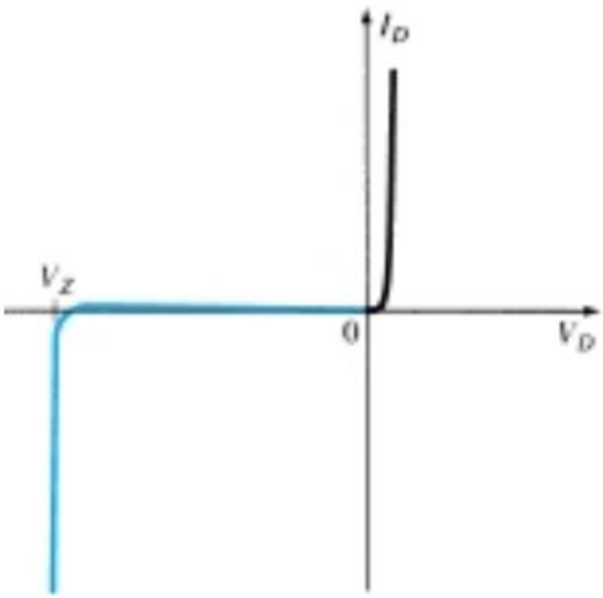


Figure 1.47 Reviewing the Zener region.

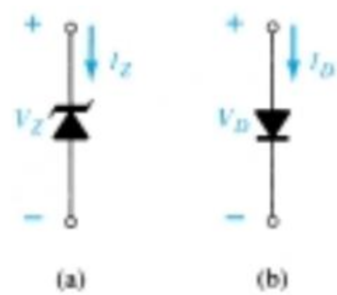


Figure 1.48 Conduction direction: (a) Zener diode; (b) semiconductor diode.

This region of unique characteristics is employed in the design of *Zener diodes,* which have the graphic symbol appearing in Fig. 1.48a. Both the semiconductor diode and zener diode are presented side by side in Fig. 1.48 to ensure that the direction of conduction of each is clearly understood together with the required polarity of the applied voltage. For the semiconductor diode the "on" state will support a current in the direction of the arrow in the symbol. For the Zener diode the direction of conduction is opposite to that of the arrow in the symbol as pointed out in the introduction to this section. Note also that the polarity of V_D and V_Z are the same as would be obtained if each were a resistive element.

1.15 LIGHT-EMITTING DIODES

As the name implies, the light-emitting diode (LED) is a diode that will give off visible light when it is energized. In any forward-biased *p-n* junction there is, within the structure and primarily close to the junction, a recombination of holes and electrons. This recombination requires that the energy possessed by the unbound free electron be transferred to another state. In all semiconductor *p-n* junctions some of this energy will be given off as heat and some in the form of photons. In silicon and germanium the greater percentage is given up in the form of heat and the emitted light is insignificant. In other materials, such as gallium arsenide phosphide (GaAsP) or gallium phosphide (GaP), the number of photons of light energy emitted is sufficient to create a very visible light source.

The process of giving off light by applying an electrical source of energy is called electroluminescence.

As shown in Fig. 1.54 with its graphic symbol, the conducting surface connected to the *p*-material is much smaller, to permit the emergence of the maximum number of photons of light energy. Note in the figure that the recombination of the injected carriers due to the forward-biased junction results in emitted light at the site of recombination. There may, of course, be some absorption of the packages of photon energy in the structure itself, but a very large percentage are able to leave, as shown in the figure.

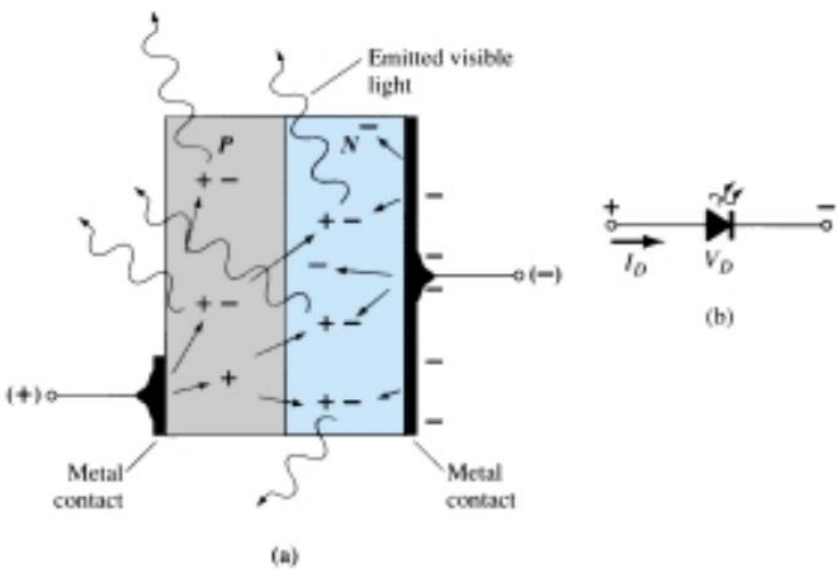


Figure 1.54 (a) Process of electroluminescence in the LED; (b) graphic symbol.

2.7 SINUSOIDAL INPUTS; HALF-WAVE RECTIFICATION

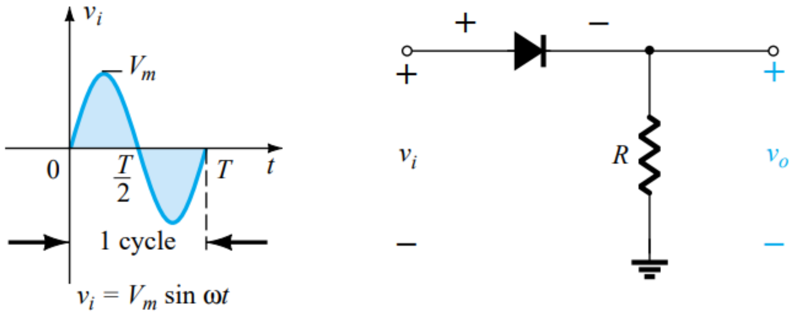


Figure 2.43 Half-wave rectifier.

Over one full cycle, defined by the period T of Fig. 2.43, the average value (the algebraic sum of the areas above and below the axis) is zero. The circuit of Fig. 2.43, called a *half-wave rectifier,* will generate a waveform v_o that will have an average value of particular, use in the ac-to-dc conversion process. When employed in the rectification process, a diode is typically referred to as a *rectifier.* Its power and current ratings are typically much higher than those of diodes employed in other applications, such as computers and communication systems.

During the interval $t = 0 \rightarrow T/2$ in Fig. 2.43 the polarity of the applied voltage v_i is such as to establish "pressure" in the direction indicated and turn on the diode with the polarity appearing above the diode. Substituting the short-circuit equivalence for the ideal diode will result in the equivalent circuit of Fig. 2.44, where it is fairly obvious that the output signal is an exact replica of the applied signal. The two terminals defining the output voltage are connected directly to the applied signal via the short-circuit equivalence of the diode.

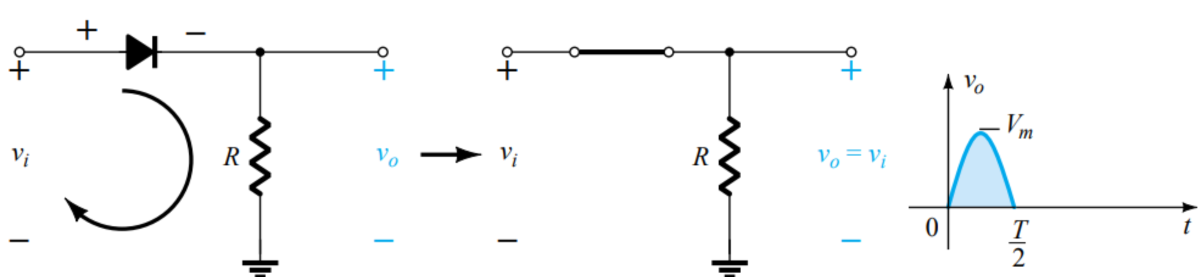


Figure 2.44 Conduction region ($0 \rightarrow T/2$).

For the period $T/2 \rightarrow T$, the polarity of the input v_i is as shown in Fig. 2.45 and the resulting polarity across the ideal diode produces an "off" state with an open-circuit equivalent. The result is the absence of a path for charge to flow and $v_o = iR = (0)R = 0$ V for the period $T/2 \rightarrow T$. The input v_i and the output v_o were sketched together in Fig. 2.46 for comparison purposes. The output signal v_o now has a net positive area above the axis over a full period and an average value determined by

$$\boxed{V_{dc} = 0.318V_m} \quad \text{half-wave} \tag{2.7}$$

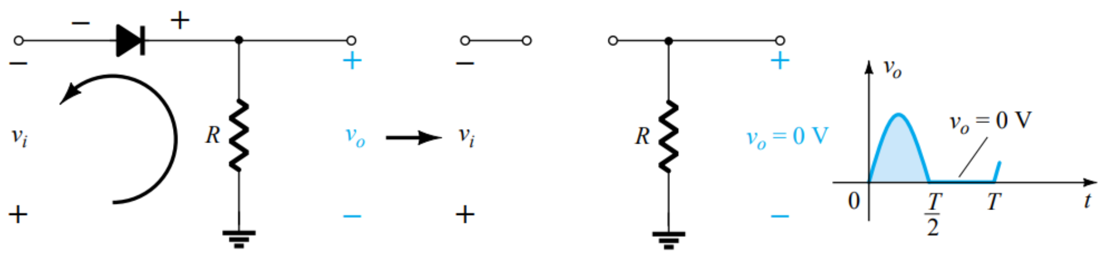


Figure 2.45 Nonconduction region ($T/2 \rightarrow T$).

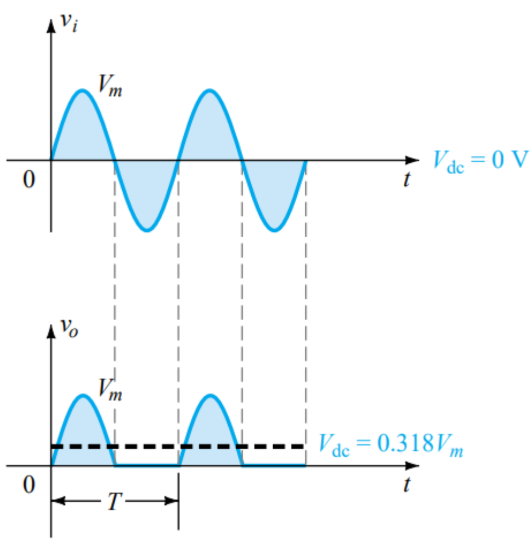


Figure 2.46 Half-wave rectified signal.

The process of removing one-half the input signal to establish a dc level is aptly called *half-wave rectification.*

$$
\begin{aligned}
V_{\mathrm{dc}} &= \frac{1}{T}\int_0^T V_{\mathrm{o}}(t)\mathrm{d}t \\
&= \frac{1}{T}\int_0^{T/2} V_m \sin(\omega t)\mathrm{d}t + \frac{1}{T}\int_{T/2}^{T} 0\ \mathrm{d}t \\
&= \frac{V_m}{T}\int_0^{T/2} \sin(\omega t)\mathrm{d}t \\
&= \frac{V_m}{T}\left[-\frac{\cos(\omega t)}{\omega}\right]_0^{T/2} \\
&= \frac{V_m}{\omega T}\{-\cos(\omega T/2) + \cos(0)\} \\
&= \frac{V_m}{\pi}.
\end{aligned}
$$

Here we have used the relation $\omega = 2\pi/T$.

2.8 FULL-WAVE RECTIFICATION

Bridge Network

The dc level obtained from a sinusoidal input can be improved 100% using a process called *full-wave rectification.* The most familiar network for performing such a function appears in Fig. 2.52 with its four diodes in a *bridge* configuration. During the period $t = 0$ to $T/2$ the polarity of the input is as shown in Fig. 2.53. The resulting polarities across the ideal diodes are also shown in Fig. 2.53 to reveal that D_2 and D_3 are conducting while D_1 and D_4 are in the "off" state. The net result is the configuration of Fig. 2.54, with its indicated current and polarity across R. Since the diodes are ideal the load voltage is $v_o = v_i$, as shown in the same figure.

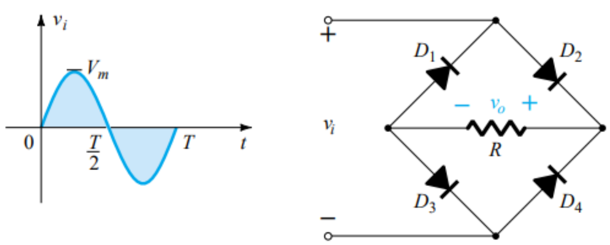


Figure 2.52 Full-wave bridge rectifier.

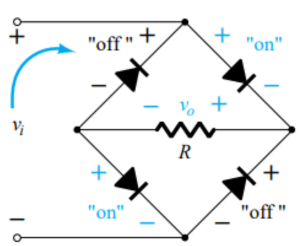


Figure 2.53 Network of Fig. 2.52 for the period $0 \rightarrow T/2$ of the input voltage v_i.

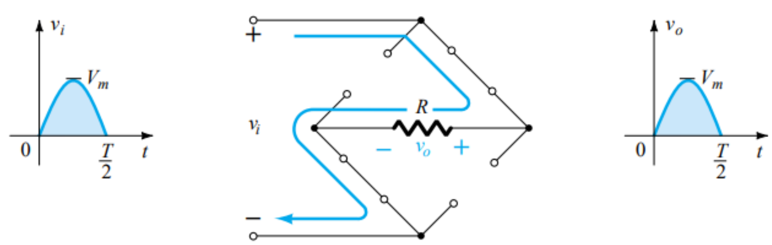


Figure 2.54 Conduction path for the positive region of v_i.

For the negative region of the input the conducting diodes are D_1 and D_4, resulting in the configuration of Fig. 2.55. The important result is that the polarity across the load resistor R is the same as in Fig. 2.53, establishing a second positive pulse, as shown in Fig. 2.55. Over one full cycle the input and output voltages will appear as shown in Fig. 2.56.

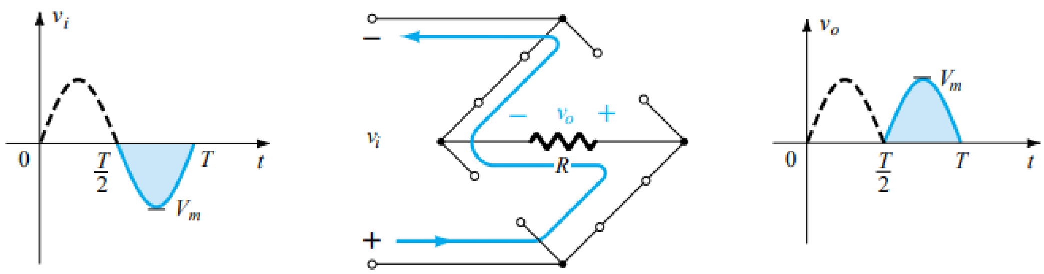


Figure 2.55 Conduction path for the negative region of v_i.

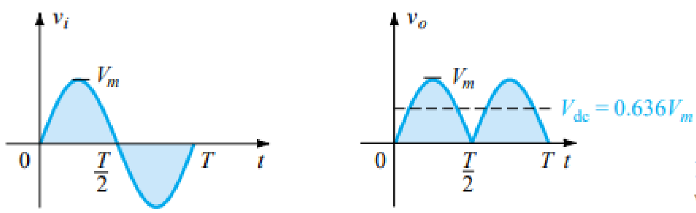


Figure 2.56 Input and output waveforms for a full-wave rectifier.

Since the area above the axis for one full cycle is now twice that obtained for a half-wave system, the dc level has also been doubled and

$$V_{dc} = 2(\text{Eq. 2.7}) = 2(0.318V_m)$$

or

$$\boxed{V_{dc} = 0.636V_m} \quad \text{full-wave} \qquad (2.10)$$

$$\begin{aligned} V_{dc} &= \frac{1}{T}\int_0^T V_o(t)dt \\ &= \frac{1}{T/2}\int_0^{T/2} V_m \sin(\omega t)dt \\ &= \frac{2V_m}{T}\int_0^{T/2} \sin(\omega t)dt \\ &= \frac{2V_m}{\pi}. \end{aligned}$$

Center-Tapped Transformer

A second popular full-wave rectifier appears in Fig. 2.59 with only two diodes but requiring a center-tapped (CT) transformer to establish the input signal across each section of the secondary of the transformer. During the positive portion of v_i applied to the primary of the transformer, the network will appear as shown in Fig. 2.60. D_1 assumes the short-circuit equivalent and D_2 the open-circuit equivalent, as determined by the secondary voltages and the resulting current directions. The output voltage appears as shown in Fig. 2.60.

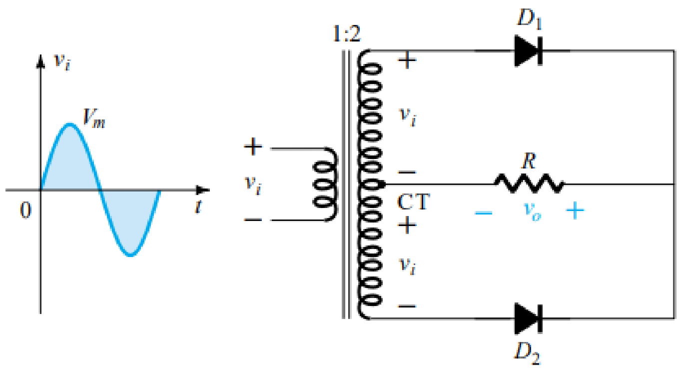


Figure 2.59 Center-tapped transformer full-wave rectifier.

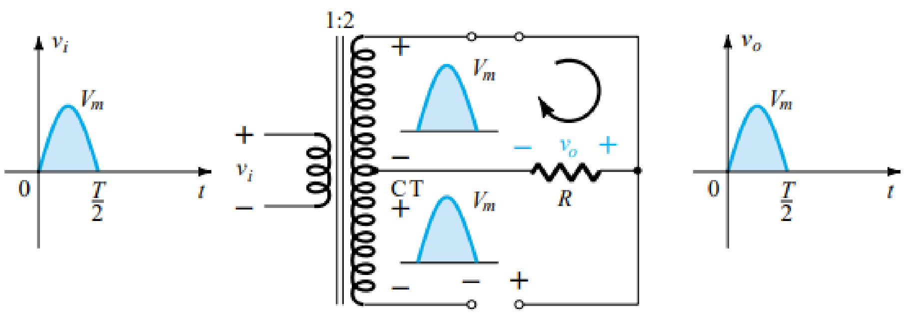


Figure 2.60 Network conditions for the positive region of v_i.

During the negative portion of the input the network appears as shown in Fig. 2.61, reversing the roles of the diodes but maintaining the same polarity for the voltage across the load resistor R. The net effect is the same output as that appearing in Fig. 2.56 with the same dc levels.

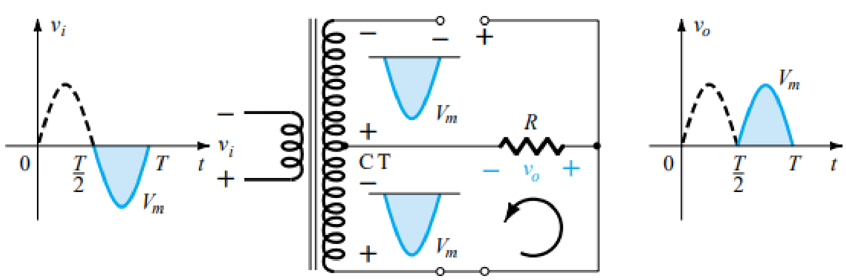


Figure 2.61 Network conditions for the negative region of v_i.